



Enhancing Atrial Fibrillation detection accuracy: A wavelet transform filtered single lead ECG signal analysis with artificial neural networks and novel feature extraction

D.U.S. Duranta^a, Md Shahin Ali^a, Abhilash Arjan Das^a, Md Mahbubur Rahman^b,
Md Manjurul Ahsan^{c,e}, Md Sipon Miah^{b,d}, Md Khairul Islam^{a,*}

^a Department of Biomedical Engineering, Islamic University, Kushtia 7003, Bangladesh¹

^b Department of Information and Communication Technology, Islamic University, Kushtia 7003, Bangladesh

^c Department of Radiology, School of Medicine, Northwestern University, Evanston, IL 60208, USA

^d Department of Signal Theory and Communications, Universidad Carlos III de Madrid, 28911 Leganés Madrid, Spain

^e Industrial & Production Engineering, Shahjalal University of Science and Technology, Sylhet 3114, Bangladesh²

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ABSTRACT

Atrial Fibrillation (AF) is a common cardiac arrhythmia that can lead to fatal outcomes. Detecting AF early is crucial for improving patient outcomes and reducing complications. However, AF is difficult to detect due to asymptomatic cases, false-positive test results, and patient non-adherence. Therefore, it is essential to develop effective methods for early AF detection that can prolong patient life. This study proposes a novel approach for AF detection using a wavelet transform-based filtered ECG signal and a neural network with a novel feature extractor. Firstly, the wavelet transform is applied to filter the ECG signal; secondly, the novel feature extractor is designed to extract important features from the filtered ECG signal. The extracted features are then used as input to a neural network for AF classification. The proposed method is evaluated on two publicly available ECG datasets, and the results show a prominent accuracy of 96% with a ROC is 0.95 and 86% with a ROC is 0.84 for the two datasets, respectively. In addition, to enhance the model's performance, we have utilized 5-fold cross-validation which achieved an accuracy of 0.99 for Dataset 1 and 0.90 for Dataset 2, respectively. Compared to existing AF detection methods, our approach provides a promising solution that can improve the early diagnosis and treatment of AF. This study presents a new avenue for AF detection and has the potential to advance the field of cardiac arrhythmia detection.

1. Introduction

Cardiovascular disease (CVD) is a significant cause of mortality worldwide, responsible for around 17.3 million deaths per year (Ahmed & Zhu, 2020). Atrial fibrillation (AF) is one of the most common heart abnormalities, affecting millions of people globally (Lippi, Sanchis-Gomar, & Cervellin, 2021). It is a cardiac arrhythmia characterized by an irregular and disorganized heartbeat caused by irregular electrical signals in the heart's atria and upper chambers. AF is more prevalent in the elderly population and can lead to an increased risk of stroke, thrombus, and other complications (Wang, Wang, & Wang, 2020). The human heart's coordinated electrical activity is vital for proper functioning, and its electrical activity is regulated by specialized cells called pacemaker cells. These cells generate electrical impulses that

travel through the heart muscle, causing it to contract and pump blood. The process of heart contraction and relaxation is known as heart rhythm. However, in AF, the heart's atria quiver or fibrillate instead of contracting and pumping blood normally, reducing blood flow efficiency and a range of symptoms, such as palpitations, shortness of breath, fatigue, and weakness (Wu, Zheng, Chu, & He, 2020). Electrocardiography (ECG) is the most common diagnostic method used to identify abnormalities in the heart (Ahmed & Zhu, 2020). It is a non-invasive and simple test that records the heart's electrical activity and produces a graph that can be analyzed to identify abnormal patterns. ECGs are a crucial tool for detecting and monitoring a wide range of cardiac conditions, including AF, the most common cardiac arrhythmia (Ghiasi, Abdollahpur, Madani, Kiani, & Ghaffari, 2017).

* Corresponding author.

E-mail addresses: duranta03@gmail.com (D.U.S. Duranta), shahinbme.iu@gmail.com (M.S. Ali), abarjan00@gmail.com (A.A. Das), mrahman@ict.iu.ac.bd (M.M. Rahman), mdmanjurulahsan@gmail.com (M.M. Ahsan), mdsiponmiah@gmail.com (M.S. Miah), khairul.ice06@gmail.com (M.K. Islam).

¹ <https://iu.ac.bd>.

² <https://sust.edu>.

However, interpreting ECG signals can be challenging due to their complexity and variability.

To improve the accuracy of AF detection from ECG signals, advanced signal processing techniques, and machine learning algorithms have been developed in recent years (Ahsan et al., 2023; Hasan, Ali, Rahman, & Islam, 2022). Machine learning (ML) approaches such as support vector machines (SVM), random forests, and logistic regression, as well as Deep Learning (DL) techniques such as artificial neural networks (ANN), convolutional neural networks (CNN), and recurrent neural networks (RNN), have become increasingly popular in the field of disease detection from ECG signals (Khriji, Fradi, Machhout, & Hossen, 2020). While ECGs effectively detect abnormalities in the heart, they cannot provide a definitive diagnosis of the underlying condition. Additional diagnostic methods such as EKG, echocardiogram, or SPECT may be required for accurate disease detection. Early detection and accurate diagnosis of AF are essential for effective treatment and prevention of associated complications, making the proposed approach in this article promising for improving patient outcomes. The accurate and early detection of AF is crucial for effectively treating and preventing associated complications (Zhao, Särkkä, & Rad, 2020). However, interpreting electrocardiography (ECG) signals can be challenging due to their complexity and variability. To address this issue, advanced signal processing techniques and ML algorithms have been developed in recent years. This article proposes a novel approach for enhancing the accuracy of AF detection by analyzing single-lead ECG signals using a combination of wavelet transform filtering, ANNs, and a new feature extraction method. The proposed approach is evaluated using a publicly available dataset of single-lead ECG signals from patients with and without AF, and results demonstrate a significant improvement in the accuracy of AF detection compared to traditional ECG analysis methods. This study contributes to the development of advanced signal processing techniques and ML algorithms for enhancing the accuracy of disease detection from ECG signals and represents a promising avenue for future research in this area.

The main contribution of this study is:

1. We propose an innovative approach for AF detection using a neural network (NN) and a novel feature extractor based on wavelet transform-based filtered ECG signals. This approach offers a unique perspective on detecting AF and provides a potential alternative to existing methods.
2. The study introduces a novel feature extractor designed specifically for the filtered ECG signal. This feature extractor aims to capture important characteristics and patterns indicative of AF, enhancing the discriminative power of the extracted features. The development of a specialized feature extractor contributes to the advancement of feature engineering techniques in AF detection.
3. The proposed method is evaluated on two publicly available ECG datasets, providing a rigorous assessment of its performance. The obtained results demonstrate a notable accuracy of 96% and 86% for the respective datasets. These results highlight the effectiveness of the proposed approach and its potential as a reliable AF detection method.

The paper is organized as follows: Section 2 provides a Literature Review of the existing research related to AF detection using ECG signals. Section 3 describes the Methodology used in this study, including the proposed AF detection model that combines wavelet transform filtering, ANNs, and a novel feature extraction method. In Section 4, we present the Results of our experiments on a publicly available dataset of single-lead ECG signals from patients with and without AF. A brief discussion of our experimental results is presented in Section 5, followed by a Conclusion and future work summarized in Section 6.

2. Related work

In recent years, there has been a growing interest in the use of ML and DL for processing biomedical signals, particularly for ECG signal processing and disease classification. AF is a common and rapidly growing cardiac arrhythmia that can be detected using ECG signal processing. The author (Liaquat et al., 2020) compared ML and DL approaches for AF detection using a single lead ECG signal. They found that DL algorithms performed more effectively than ML methods, achieving an accuracy of approximately 86%, although they did not use feature extraction techniques. The author (Xiong, Stiles, & Zhao, 2017) used a 16-layer 1-dimensional convolutional neural network-based model with skip connections to classify ECG signals, including AF. They worked on a single dataset and achieved an accuracy of 90%. These studies demonstrate the potential of ML and DL approaches for improving the accuracy of AF detection from ECG signals. In (Wang et al., 2020), the authors proposed an ANN-based model for AF detection, where wavelet packet transform and correlation function was used as a feature extractor to extract features from the ECG signal. By using the feature extractor, they achieved high accuracy.

Similarly, the author (Fang et al., 2021) developed a neural network-based model for AF detection, which utilized a dual-channel neural network and applied noise reduction to the ECG signal twice. Their model achieved an F1 score of 83% and 90% for AF and normal signals, respectively. In (Khriji et al., 2020), the authors employed DL for AF detection. They also used a simple neural network to detect AF in the MIT-BIH arrhythmia dataset, achieving an accuracy of 93%. They also used a feature extraction and medical knowledge-based algorithm for AF detection. The author (Ahmed & Zhu, 2020) developed a heart monitoring system that can detect AF in its early stage using their medical knowledge. They analyzed the single lead ECG signal and extracted features such as RR interval and heartbeat. They then developed an algorithm to analyze these features and detect AF. In recent times, DL methods such as CNN and RNN have gained popularity in signal processing and classification. The author (Taji, Chan, & Shirmohammadi, 2017) proposed a novel model to reduce False Alarm (FA) rates during AF detection due to poor-quality ECG signals. They used a Deep Belief Network (DBN) and found that their model increased the accuracy of AF detection. The combination of CNN and LSTM is widely used in biomedical signal classification. The author (Wang & Li, 2020) proposed an AF detection model using a cascade of CNN and LSTM. They found that the cascade of these two DL methods performed better than their concatenation, achieving an F1 score of 82%.

The author (Cheng et al., 2020) proposed a CAD-based model to detect AF. They used time-frequency analysis in conjunction with the DL method. They divided their method into two parts. In the first part, a time-frequency chromatograph is created from the data, and in the second part, this chromatograph is used as input to the hybrid model developed with LSTM and CNN. They got a high accuracy but they used photoplethysmography (PPG) which is not a common diagnostic method as ECG. In another work, The author (Petmezas et al., 2021) developed a CNN and LSTM-based hybrid model to detect AF from ECG signals. The CNN is used as a feature extractor and then this feature is used in the LSTM model to classify the condition. This model can utilize focal loss, a type of improved cross-entropy to deal with training data imbalance. In (Rao, Kolekar, & Martis, 2021), the authors performed a frequency domain analysis of single-lead ECG signal using Welch, Auto Regression (AR), and Lomb scargle periodogram methods. They made a comparison between deep neural networks (Temporal Convolution Network (TCN) and Deep Convolutional Neural Network (DCNN)) and traditional ML (Decision Tree (DT) and Support Vector Machine (SVM)) approach. And They discovered that DL gives over performance than ML.

The author (Tran, Li, Nocera, Shahabi, & Xiong, 2020) used a pre-trained convolutional neural network-based algorithm named MultiFusionNet to detect AF from ECG signal. This method uses a multiplicative fusion method which actually combines two deep neural

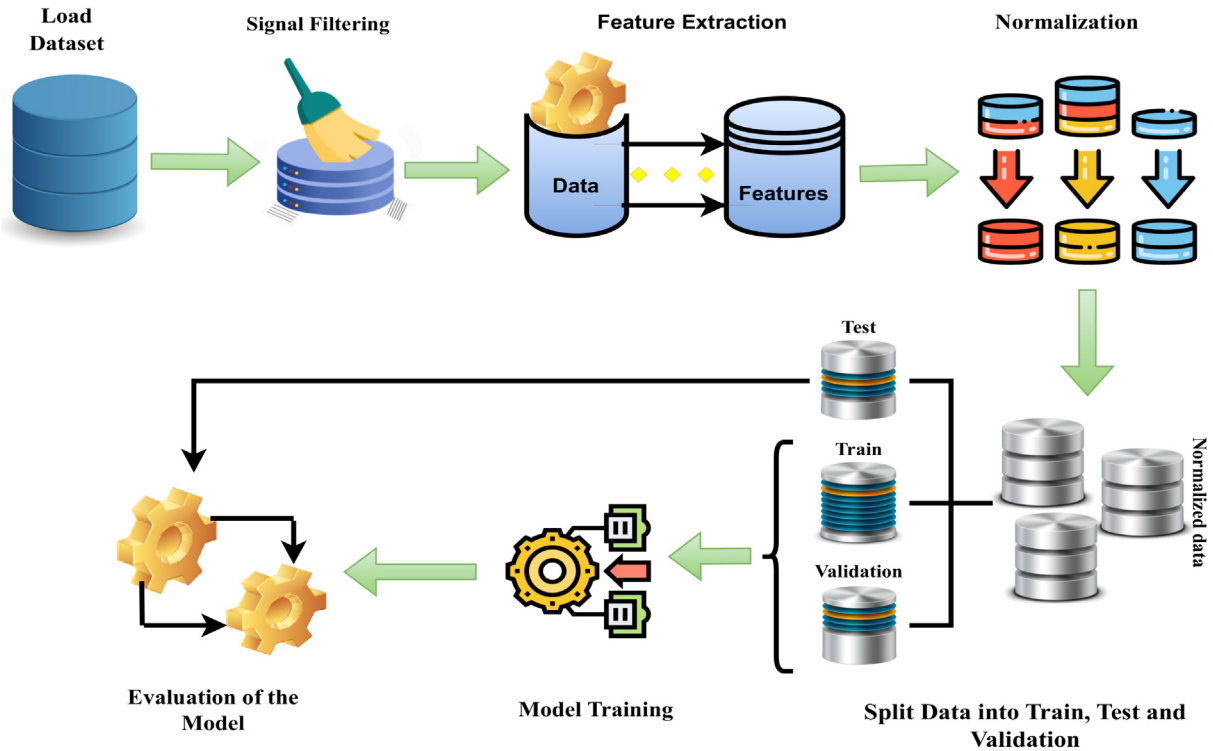


Fig. 1. Flow diagram of the proposed methodology.

networks which are trained on different sources of knowledge. The relevant feature that was retrieved can be used by MultiFusionNet to enhance the performance of the DL model. They got quite good accuracy in their work. The author (Ivanovic, Atanasoski, Shvilkin, Hadzievski, & Maluckov, 2019) proposed a deep neural network has been in which a CNN and RNN are used to extract features from the RR interval of ECG and these features are used in long-term ECG signals for the classification of AF and Atrial Flutter. They found around 89% accuracy in testing. In the last decade of the 19th century, some CAD-based AF detection-related work had performed also. Such as The author (Artis, Mark, & Moody, 1991) proposed an ANN-based AF detection model which also can be used to detect the pattern from the ECG signal. They found a good accuracy of 92%. A transfer learning-based neural network to classify cardiac arrhythmia has been proposed by the author (Isin & Ozdalili, 2017). They took a Transferred deep convolutional neural network name AlexNet to bring out the feature and put this feature in a simple neural network to classify three cardiac conditions. They got a high testing accuracy of 92%. In (Parvaneh, Rubin, Babaeizadeh, & Xu-Wilson, 2019), the authors reviewed many research papers which are worked on cardiac arrhythmia detection using a DL algorithm. They analyze those papers with five aspects and these are utilized dataset, application, type of input data, model architecture, and performance evaluation. They also compare the DL and ML approaches for ECG signal classification.

In this field, also ML algorithms-based work has been performed. Support vector machine, random forest, Decision tree, linear regression, etc. algorithms are used for this purpose. The author (Asgari, Mehrnia, & Moussavi, 2015) proposed an AF episode detection model using an ML algorithm and signal processing. They use wavelet transform and support vector machine(SVM) algorithms. This method does not depend on R-peak, p-peak, and heartbeat detection. they got satisfactory accuracy. The author (Kalidas & Tamil, 2019) proposed a fully automated AF episode detection model using discrete-state Markov models with Random Forests algorithm from ECG signal. Discrete-state Markov models are used in this work to detect RR interval and other

statistical features from ECG signals, and these features are used to classify AF with Random Forests Algorithm. They worked with MIT-BIH AF Database (AFDB) and the MIT-BIH Arrhythmia Database (MITDB).

In conclusion, various DL and ML methods have been proposed for the detection of AF from ECG signals. These methods include the use of (CNN), RNNs, deep belief networks, and hybrid models that combine CNNs and RNNs. In addition, some studies have also used signal-processing techniques and medical knowledge-based algorithms for feature extraction and AF detection. Overall, these methods have shown promising results in detecting AF from ECG signals with high accuracy, and they have the potential to significantly improve the diagnosis and treatment of this cardiac condition. However, further studies are needed to validate these models on larger datasets and in clinical settings.

3. Materials and methods

The following section will detail the methodology that we have proposed to employ. Fig. 1 shows the conceptual structure of our overall working procedure.

3.1. ECG signal data acquisition

3.1.1. Dataset 1

In this work, we have used two very popular and publicly available single-lead ECG datasets. The first one was collected from a website named Mendeley Data (Pławiak, 2018). The ECG signals have collected from 45 patients. The dataset is classified into 17 categories of CVD. The ECG signal is collected at a sampling frequency of 360 and the time of the signal is 10 s and the total data is 1000. But this dataset is too much unbalanced and classified into too many classes. For this reason, we modified the dataset and made a new dataset with normal sinus rhythm and AF data, which contains around 411 data.

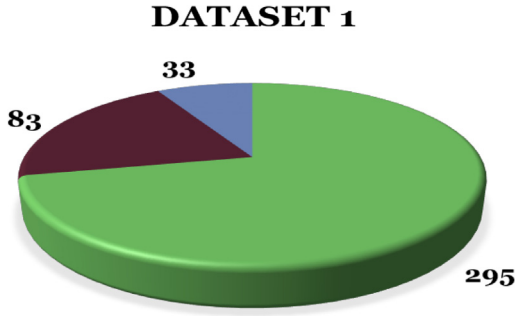


Fig. 2. Dataset 1 distribution in train test and validation.

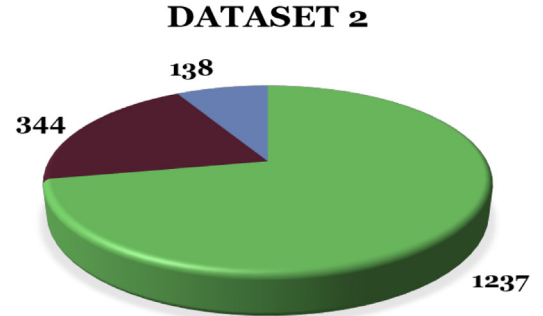


Fig. 3. Dataset 2 distribution in train test and validation.

3.1.2. Dataset 2

Dataset 2 has been collected from ‘The PhysioNet in Cardiology Challenge 2017’ (Clifford et al., 2017). The dataset contains 8528 single-lead ECG data. Among this data 5154 data are normal, 771 are AF, 2557 data are other rhythm and 46 are noisy rhythm. The sampling frequency of this dataset is 300. The time fragment of this data is not specific. The time fragment range of this data is from 9 s to just over 60 s. This dataset is also a unbalance dataset and that is why the data has been modified and made a dataset of 771 AF data and randomly selected 948 normal data and made two classes with this data. The modified datasets are available publicly on the following GitHub link <https://github.com/Duranta001/AF-detection-database>.

The dataset has been partitioned into three subsets: a training set consisting of 80% of the data, a testing set consisting of 20% of the data, and a validation set consisting of 10% of the training set. A representative sample of the dataset is illustrated in the accompanying Fig. 4.

3.2. Wavelet transform

Wavelet transform is a common and popular mathematical tool in the field of signal processing, mostly for time–frequency analysis of signals. Wavelets are small, localized signals that can be used to represent more complex signals in a compact form. The wavelet transform represents signals and data at several resolutions, giving both time and frequency information in a single representation. The basic operation of the wavelet transform involves transforming a signal or image into a representation in terms of wavelets (Sahambi, Tandon, & Bhatt, 1997). This is typically accomplished using a multi-step process:

Decomposition: The original signal or image is decomposed into different frequency components using a wavelet transform algorithm. This is typically done by convolving the signal or image with wavelets at various scales and locations.

Subsampling: The decomposed signal or image is then subsampled to produce a series of wavelet coefficients. These coefficients represent the relative contributions of the wavelets at different scales and locations to the original signal or image.

Thresholding : The wavelet coefficients are then thresholded to remove any coefficients that are below a certain magnitude. This helps to reduce the size of the representation and remove any noise or other unwanted components from the signal or image.

Reconstruction: The wavelet coefficients are then used to reconstruct the original signal or image. This is typically done by convolving the coefficients with the original wavelets and summing the results.

The wavelet transform of a function or signal $f(t)$ is typically represented mathematically as the convolution of the function with a wavelet function $\Psi(t)$:

$$W(a, b) = \int_{-\infty}^{\infty} f(t) \Psi_{a,b}(t) dt \quad (1)$$

where ‘a’ is a scaling factor that determines the scale of the wavelet, and ‘b’ is a translation factor that defines the wavelet’s position. The wavelet function $\Psi_{a,b}(t)$ is defined as:

$$\Psi_{(a,b)}(t) = \frac{1}{\sqrt{a}} \Psi\left(\frac{t-b}{a}\right) \quad (2)$$

where $\Psi(t)$ is the mother wavelet, a scaling factor that defines the shape of the wavelet. The inverse wavelet transform is given by:

$$f(t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} W(a, b) \Psi_{a,b}(t) dt \quad (3)$$

This representation can be used to reconstruct the original signal or image from the wavelet coefficients.

3.3. Data preprocessing

Pre-processing is a data preparation step in ML and data analysis where raw data is transformed into a clean and structured format that is suitable for modeling and analysis (Ali et al., 2023; Islam, Ali, Ali et al., 2021). This process involves several tasks such as data cleaning, feature scaling (normalization or standardization), feature selection, and data transformation (encoding categorical variables, transforming variables) (Ali, Miah, Haque, Rahman and Islam, 2021; Islam, Ali, Miah et al., 2021). In this work, data preprocessing has been performed in two stages. The first stage is performed before feature extraction and the other is performed after the feature extraction. In the first stage, the signals have filtered by using a wavelet-based filter where the high-frequency noise is removed from the signal. In the other stage, the extracted features are preprocessed for model input. At this time, the extracted features are converted into a numpy array and the scaling technique have performed.

The signal has filtered before passing into the feature extraction function. In this step, we deconstruct the signal by performing a four-level wavelet transform with a ‘sym4’ wavelet. This wavelet transform deconstructs the signal into one approximate coefficient and 4 detailed coefficients according to frequency. The detailed coefficient of level one and level 2 have converted into zero. Though these two coefficients contain high-frequency data so when it became zero, the high-frequency noise is removed. Then the signal is reconstructed by using all these coefficients. The mathematical representation of the ‘sym4’ wavelet is,

$$h(t) = (1 + t) \cdot e^{-\frac{t^2}{2}} \cdot [t \geq -1] \cdot [t < 1] \quad (4)$$

where $h(t)$ is the wavelet function and t is the time variable. Fig. 5 shows the original and filtered signal.

3.4. Feature extraction

A feature is a distinct attribute or aspect of something, which serves as a characteristic that helps to identify it or distinguish it from others.

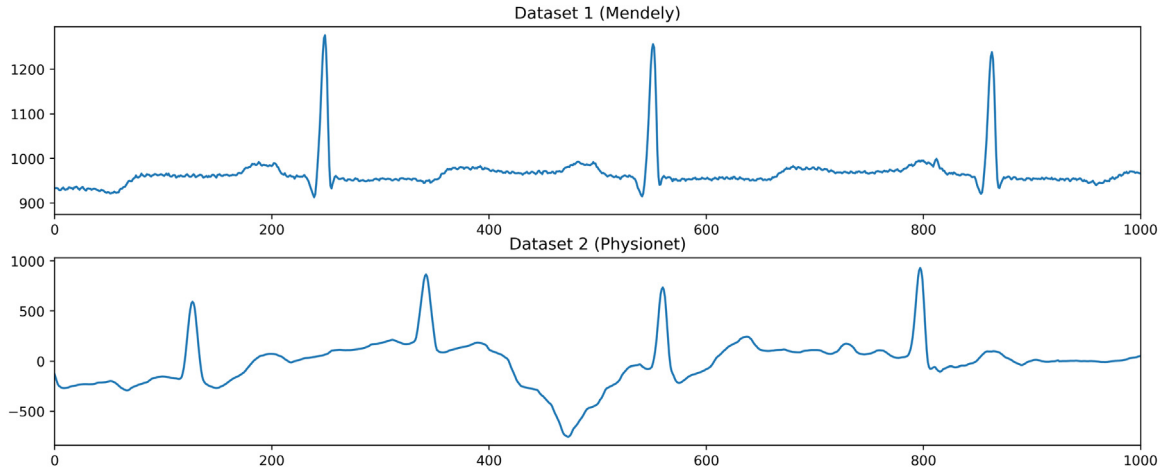


Fig. 4. Sample of ECG signal Datasets.

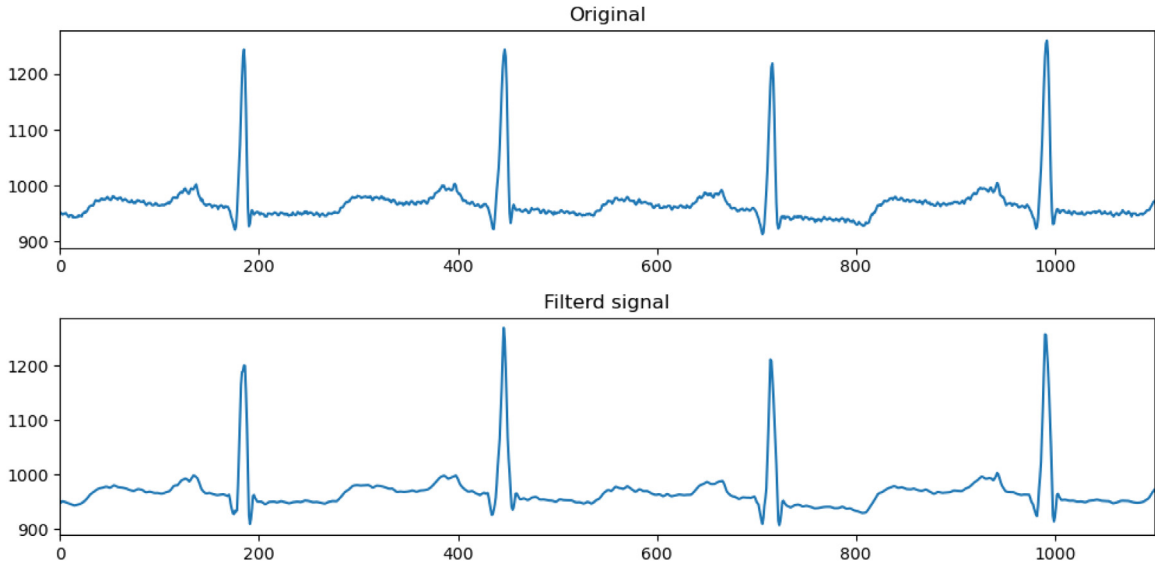


Fig. 5. Original and filtered signal.

The purpose of feature extraction is to find and extract the most significant and pertinent information from the data, and to represent it in a way that a machine-learning algorithm can easily understand and use. The features to extract are determined by the specific challenge and data type, and the process of feature extraction can be quite complex and time-consuming (Ali, Islam et al., 2021; Islam, Ali, Das, Duranta, & Alam, 2020). In our work, we developed and used a novel feature extractor that can extract features from single-lead ECG signals. By using this feature extractor, we can get 10 highlighted features of the ECG signal (see Fig. 6). The extracted features are:

Heart Beat: A heartbeat is one of the common features of heart disease detection. we detect the heartbeat from the ECG signal by using the R peak. Firstly, we have done 4 level wavelet transform to the signal and replaced the 1st, 2nd-level detailed, and 4th-level approximate coefficient with zero. By doing this, the R peaks are highlighted, and then we use a peak detection function that we get from the Python module to detect the r peak. Then we count the R peak number and time. Then we performed a mathematical calculation to detect the heartbeat.

The mathematical equation for heartbeat detection is,

$$Heartbeat = \frac{N \times 60}{t} \quad (5)$$

where t is the ECG recording period and N indicates the number of R peaks.

R peak Height: When we detect the R peak, the module provides us with the R peak height also. Basically, this height is the amplitude of the R peak. We take the maximum minimum and average value of R peak height as the required feature.

The mathematical equation for calculating the average of R peak height is,

$$R_{avg} = \frac{1}{n} \sum_{n=0}^n F(n) \quad (6)$$

where R_{avg} is the mean of R peak height, $F(n)$ is the list of R peaks height and n is the length of the list.

RR interval: RR interval is too much important in the case of detecting discontinuity and abnormality of the heart. We performed a basic subtraction to detect the RR interval. In the previous, we got all the R peaks with their position and amplitude. We subtract all the R peak positions from one to another sequentially and get the RR interval. We take the maximum, minimum, and average RR interval from every signal as a feature.

$$F(R_{int}) = [R_{n+1} - R_n]_{n=0}^{n=l} \quad (7)$$

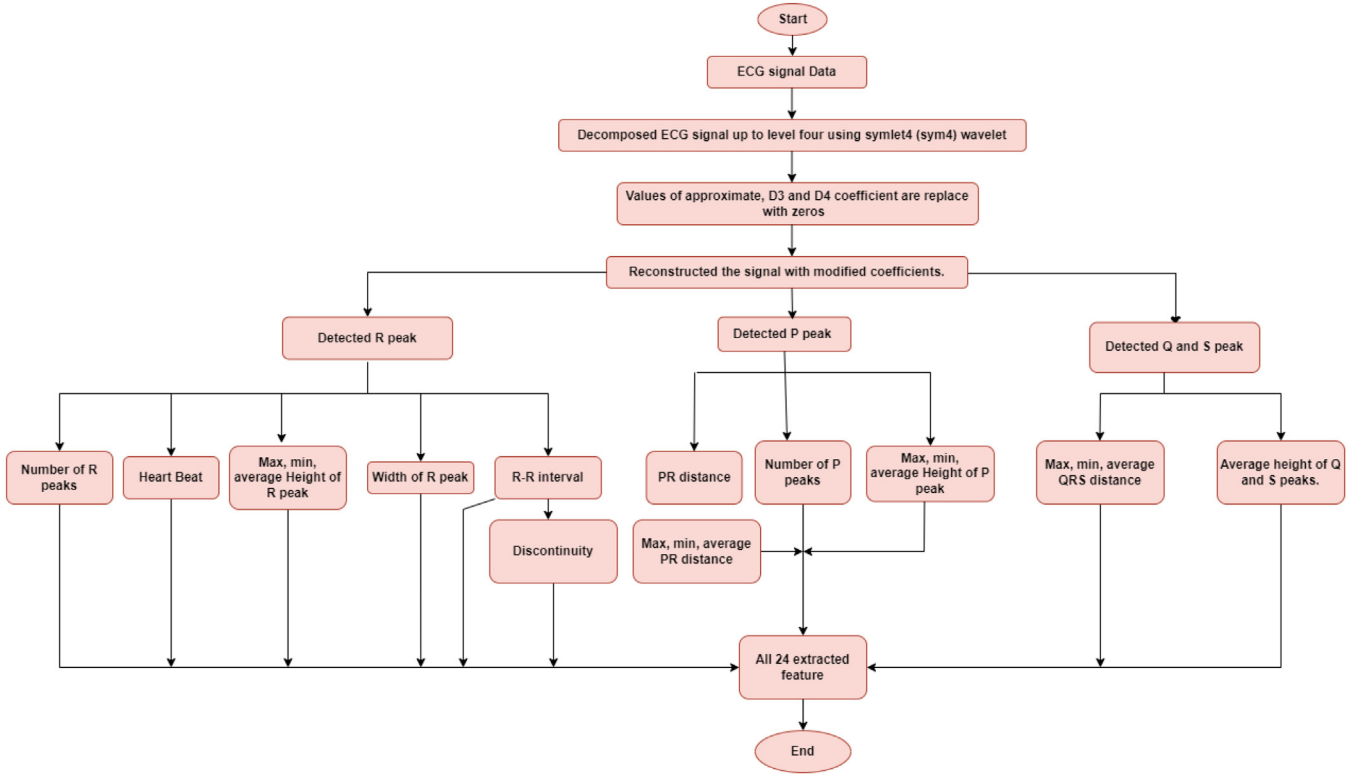


Fig. 6. Flow-diagram of feature extraction function.

where R_n is the first r peak position and R_{n+1} is the R peak position which is positioned immediately after the R_n . and l is the total number of r peaks.

$$RR_{intavg} = \frac{1}{n} \sum_{n=0}^n F(R_{int}) \quad (8)$$

where RR_{intavg} is the mean of the RR interval, $F(R_{int})$ is the RR interval and n is the length of the RR interval list.

Discontinuity: We detect the discontinuity by comparing the RR interval and the normal range of the RR interval. RR intervals typically last between 0.6 and 1.2 s. If any of the RR intervals of the ECG signal is out of the normal range, then we marked the signal as having a discontinuity.

R Peak width: R peak width is the width measured in the middle of the QRS wave. This width has been detected by using the Python module. We provide the R peak position and amplitude into the module function and get the width of the QRS wave at the middle height. We take the maximum, minimum, and average values of R wave width as a feature.

P peak height: We detect the P peak using the same module used in R peak detection. By using the R peak position, we get the P peak. And at this time we get the P peak height. We take the maximum, average, and minimum values of the P peak height.

PR Distance: PR distance is the time distance between P and R peaks which are placed sequentially. The average PR interval is between 0.12 and 0.20 s. We simply subtract the positions of P and R which are placed sequentially one after another. Then the subtract value is converted into a time value by using the sampling rate. We take the maximum, minimum, and average values of the PR interval.

QRS interval: QRS interval is the time of occurring QRS wave. QRS wave is also known as QRS complex. For the QRS interval, we detect the Q and S peaks. For this, we complemented the signal by multiplying the signal with -1. Then we detect the Q and S peaks and calculated the distance between Q and S peak positions. This Distance is the QRS

interval. We take the maximum, minimum, and average values from every signal.

The number of P and R peaks: We also take the total number of P and R peaks of a signal as a feature.

Q and S peak height: At the time of Q and S peak detection, we got the Q and S peak height or amplitude. We take only the average value of all the Q and S peak heights of a signal.

3.5. Normalization of data

In machine and DL, data normalization refers to the process of transforming data so that it is in a standard scale and format, making it easier for algorithms to process and learn from it. There are numerous varieties of data normalizing methods (Saranya & Manikandan, 2013), including min-max normalization, log normalization, decimal scaling normalization, normalization to unit length, and z-score normalization. Among all of these techniques, we have used Z-score normalization which is also known as standardization. Standardization changes data so that it has a mean of zero and a standard deviation of one. And for 1D mathematical data, this technique performs at a top-notch level. The transformed data has a conventional normal distribution with a mean of zero and a standard deviation of one. Standardization is done by removing the data mean from each observation and dividing the result by the standard deviation. The modified data will have the same shape as the original data, but it will have a mean of zero and a standard deviation of one. The mathematical equation for standardization (Z-score normalization) is:

$$Z = \frac{X - \mu}{\sigma} \quad (9)$$

where Z is the standardized value, X is the original value, μ is the mean of the data, and σ is the standard deviation of the data.

Algorithm 1 Algorithm for extracting features from ECG signal.**Input:** Raw data (e.g., ECG signal)

Sampling rate

The normal range for RR interval

Module for detecting peaks and widths

Outputs:

Extracted features (e.g., heartbeat, R peak height, RR interval, etc.)

- 1: Perform 4-level wavelet transform on the raw ECG signal.
- 2: Replace the 1st, 2nd-level detailed, and 4th-level approximate coefficients with zero to highlight the R peaks.
- 3: Use a peak detection function to detect the R peaks.
- 4: Count the number of R peaks and calculate the time between each R peak to detect the heartbeat.
- 5: Calculate the maximum, minimum, and average height of R peaks to extract the R peak height feature.
- 6: Subtract the positions of sequential R peaks to calculate the RR interval.
- 7: Calculate the maximum, minimum, and average values of the RR interval to extract the RR interval feature.
- 8: Compare the RR interval with the normal range and mark the signal as having a discontinuity if any interval is out of range.
- 9: Use the module to detect the width of the QRS wave at the middle height to extract the R peak width feature.
- 10: Use the same module to detect the P peak and extract the maximum, average, and minimum values of the P peak height feature.
- 11: Subtract the positions of sequential P and R peaks and convert the subtract value into a time value by using the sampling rate to extract the PR distance feature.
- 12: Complement the signal by multiplying the signal with -1 to detect the Q and S peaks for QRS interval extraction.
- 13: Calculate the distance between Q and S peak positions to extract the QRS interval feature.
- 14: Count the total number of P and R peaks to extract the number of P and R peaks feature.
- 15: Return the extracted features as the output.

3.6. Proposed artificial neural network model for AF detection

We designed an ANN architecture specifically for AF detection. The model comprises five fully connected layers including input layer, three hidden layer and the output layer. The input layer consists of 24 nodes, followed by a hidden layer with 15 nodes and a ReLU activation function. Another hidden layer with 15 neurons and a ReLU activation function is subsequently added. The final hidden layer has 10 neurons with ReLU activation. To prevent overfitting, a dropout layer with a rate of 0.25 is incorporated. The output layer consists of two nodes with a sigmoid activation function, which provides the probability of AF. The model is trained using sparse categorical cross-entropy loss and the Adam optimizer, with a learning rate set at 0.001.

Our AF detection model uses a novel feature extraction method that involves extracting 24 important features from each ECG signal using a dedicated feature extractor. These features are then compiled into a list that contains the extracted features of all ECG signals. Before using the features in the ANN model architecture, we normalize them to ensure that they have a consistent scale. The normalized features are then fed into the input layer of the ANN model. By extracting features separately from the ANN model architecture, we can ensure that our model can

handle a wide range of ECG signals, including those with complex characteristics. This approach also allows us to fine-tune the feature extraction process independently of the ANN model, which can improve the accuracy of our AF detection algorithm. Training is conducted with careful consideration of overfitting, ensuring the generalizability and robustness of the model. Overall, this proposed ANN model is designed to effectively capture the distinctive features relevant to AF detection, enabling accurate classification of AF cases. The incorporation of dropout regularization enhances the model's ability to generalize to unseen data, ensuring its reliability and robustness in practical applications. Table 2 represents the parameters and hyperparameters of our proposed model.

The operation of ANN is inspired by the structure and operation of the human brain (Abdalla & Esmail, 2018). It is composed of several connected processing nodes (neurons) that communicate via weighted connections to process and transmit data. ANN are taught to execute a specific function by modifying the weights of the connections between neurons, through the use of input-output sample data (Jain, Mao, & Mohiuddin, 1996). The information is received by the input layer, and the final result is provided by the output layer. The hidden layers process the information, and the connections between the neurons are called synapses. The strength of these connections, or weights, determines how the information is processed.

The training process is the key to the success of an ANN. ANN works in two steps including feed-forward and back-propagation. When the training start, the feed-forward step is performed first. Feed-forward is the method of passing the input data through the network to produce an output. The input is routed through the input layer first, then through each hidden layer in turn until it reaches the output layer. Each neuron processes the input at each layer, applying a weighted sum and an activation function to create an output that serves as the input for the following layer. the output of the hidden layer can be calculated as follows:

$$Z = W \times X + b \quad (10)$$

where Z is an $n \times 1$ column vector representing the activation of the n hidden neurons, X is an $m \times 1$ column vector representing the activation of the m input neurons, W is an $n \times m$ weight matrix, and b is an $n \times 1$ bias vector. The activations of the hidden layer neurons are then passed through an activation function to obtain the final outputs of the hidden layer, denoted as a:

$$a = g(Z) \quad (11)$$

where g is the activation function, which can be any type of non-linear function such as the sigmoid function, hyperbolic tangent function, or rectified linear unit (ReLU).

Then, the back-propagation step starts to perform to tune the parameters and biases in such a way that the loss function gradients concerning the weights and biases are minimized. The gradients are computed using the calculus chain rule and propagated backwards through the network toward the input layer from the output layer. ANNs carry out the primary operations at their input, hidden, activation, and output layers. A table of a list of parameters and hyperparameters in ANN is shown in the given Table 1.

Fig. 7 shows the architecture of our proposed model.

3.7. Training and performance

The environment in which a model is trained is extremely important in research, as it can impact the effectiveness of the model. Our proposed model is implemented using Python 3 and the Keras framework. It is implemented in a Google Colab GPU environment with 12 GB RAM and Jupyter Notebook in a system with Intel(R) Core(TM) i5-8250U CPU processor and 4 GB RAM. We have also utilized the Python3 Keras library, with TensorFlow as the backend.

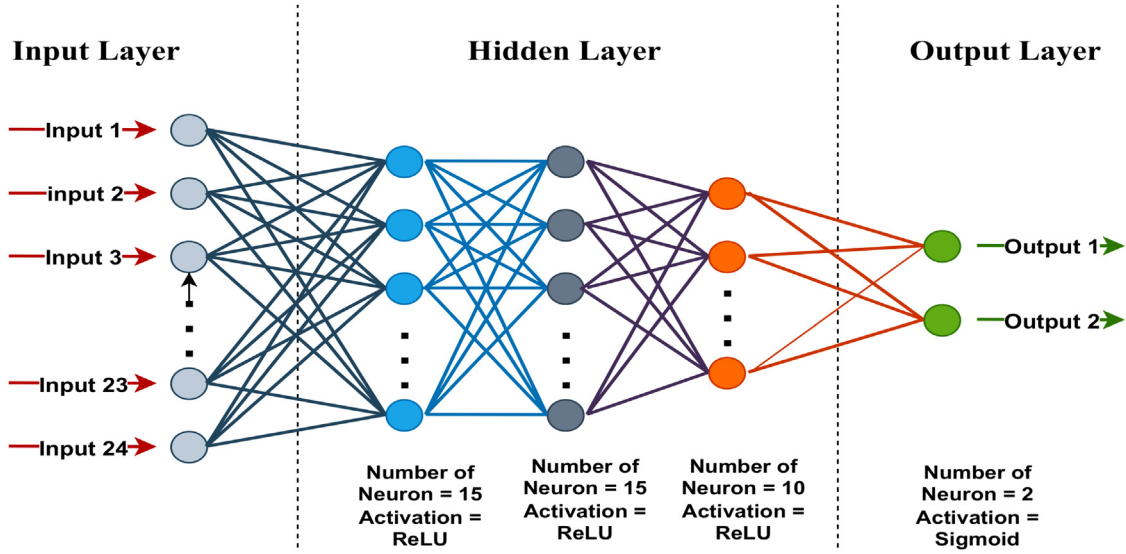


Fig. 7. Architecture of our proposed ANN model for AF detection.

Table 1

A list of parameters and hyperparameters in a ANN.

Layer	Parameters	Hyperparameters
Input layer	None	Number of input features
Hidden layers	Weights and biases of each neuron	Quantity of hidden layers, Dropout rate, Quantity of neurons per layer, and Activation function.
Output layer	Weights and biases of each neuron	Quantity of output classes, Quantity of neurons per layer, Activation function

Table 2

List of parameters and hyperparameters of our proposed model.

Hyperparameters	Measurements and techniques
Layers	5
Activation function	ReLU
Classification activation function	Sigmoid
Optimizer	Adam
Loss function	SparseCategoricalCrossentropy

3.8. Model training

To make accurate predictions, an ANN is trained by using data to update its parameter. The aim of training an ANN is to find the best combination of weights and biases that minimizes the difference between the network's outputs and the target values. To get a satisfactory performance, we fine-tuned the model parameters. The important parameters which are used in this experiment with their values are mentioned below:

- **Activation Function(ReLU):** It is a simple mathematical function that takes an input value and returns the maximum of that value and 0. In other words, if the input is positive, ReLU returns the input value, but if the input is negative, ReLU returns 0. In the hidden layer of our model, we used this activation function.
- **Activation Function(Sigmoid):** In neural networks, the sigmoid activation function is frequently employed. It converts any real number to a number between 0 and 1, representing the probability of a certain event occurring. In neural networks, the sigmoid activation function is frequently employed. It converts any real number to a number between 0 and 1, representing the probability of a certain event occurring.

- **Loss Function(SparseCategoricalCrossentropy):** In our study, we employed SparseCategoricalCrossentropy as a loss function to estimate the loss. It is a variation of categorical cross-entropy used when the target variable is sparse, meaning that it is represented by a single integer label for each observation rather than a one-hot encoded vector.
- **Optimizer(Adam):** Adam (Adaptive Moment Estimation) is a prominent optimization technique used to update model parameters in DL and ML. It is a stochastic gradient descent (SGD) optimization algorithm extension that combines the concepts of momentum and learning rate adaptation.
- **Epoch(50/20):** We utilize 50 and 20 epochs for Dataset 1 and Dataset 2, respectively. An epoch in ML is a complete iteration over the entire training dataset. In other words, an epoch represents one pass of the algorithm over all of the training examples. During each epoch, the algorithm updates the model parameters based on the current set of weights, biases, and other parameters.

The proposed model was developed and optimized on the extracted features which are extracted from the single-lead ECG signal dataset to identify the AF issues. We divided our dataset into training, validation, and testing. The model was trained on 80% of the data, validated on 10% taking from the training set, and tested on the remaining 20% to evaluate its performance. In addition, we tuned the model to improve the performance of AF detection.

3.9. 5-Fold cross-validation

To evaluate the performance of our proposed detection model, we employed the 5-fold cross-validation technique, which is a widely used method for assessing the effectiveness of machine learning models. The dataset was randomly divided into 5 equal-sized folds, and the model was trained and evaluated 5 times, with each fold serving as the validation set exactly once. This method allows us to obtain a more accurate estimate of the model's generalization ability and mitigates the risk of overfitting. Using 5-fold cross-validation, we ensure that our proposed detection model performs well on unseen data and provides reliable results.

3.10. Performance metrics

Performance metrics are essential for evaluating the effectiveness, efficiency, and quality of a system, model, or process. Different performance metrics serve various purposes and are used to evaluate

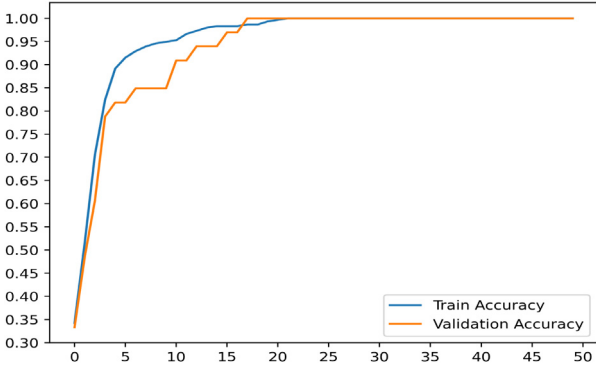


Fig. 8. Training and validation accuracy curve of proposed AF detection ANN model for dataset 1.

models in different domains. Commonly used performance metrics include accuracy, precision, recall, F1-score, AUC (area under the curve), mean squared error (MSE), root mean squared error (RMSE), etc. The selection of a performance metric depends on the type of problem, evaluation criteria, and the desired outcome. In evaluating a classification model, a Confusion Matrix is often used, which includes four important terms: true positive (TP), true negative (TN), false positive (FP), and false negative (FN).

The model's performance is assessed using different metrics, which are computed using the equation given below:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (12)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (13)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (14)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

The value of all performance metrics ranges from 0 to 1.

4. Results

The validation of our AF detection model was performed using single-lead ECG (Electrocardiogram) signal data obtained from both normal and AF patients. Initially, we conducted several experiments to optimize the performance of the model by adjusting various parameters, such as the number of neurons, the number of layers, the loss function, and the number of epochs. During the model's training process, we used the Adam optimizer with a learning rate of 0.001. The SparseCategoricalCrossentropy loss function was employed to compute the loss during training. During our research, we utilized a total of 50 epochs in the model training process. All 1267 trainable parameters were used for training the model. For the output layer of the model, we employed the sigmoid activation function, which is suitable for binary classification tasks like the one performed by our proposed model.

Our proposed model in this study was developed using two distinct datasets of ECG signals. In Dataset 1, we used a total of 411 ECG signal data, out of which 295 signals were utilized for training the model, and the remaining 116 signals were used for both model validation and testing. Details of the data used for training, validation, and testing are provided in Fig. 2. Similarly, in Dataset 2, we utilized 1237 ECG signals for training the model, and out of 1719 ECG signals, we used 482 images for both model validation and testing (see Fig. 3). We evaluated the performance of our proposed AF detection model on these two datasets separately. Initially, we trained and evaluated the performance of our model using Dataset 1. Subsequently, we repeated the same process using Dataset 2. Based on the results obtained from

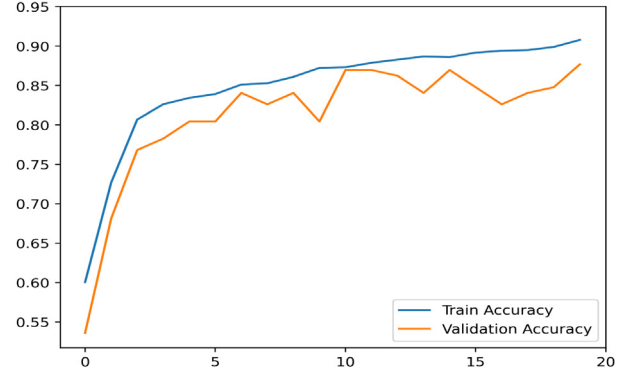


Fig. 9. Training and validation accuracy curve of proposed AF detection ANN model for dataset 2.

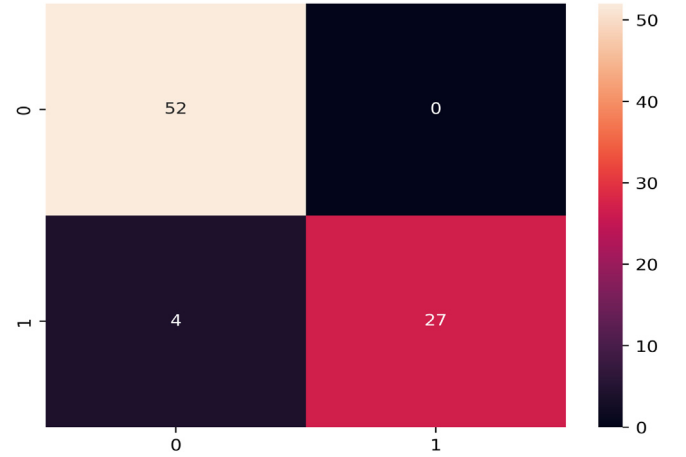


Fig. 10. Confusion metrics for Dataset 1 (0 = 'Normal', 1 = 'AF').

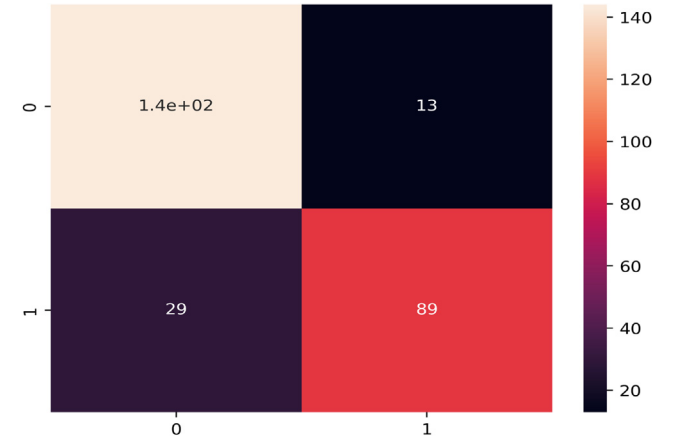


Fig. 11. Confusion metrics for Dataset 2 (0 = 'Normal', 1 = 'AF').

the confusion matrix of Dataset 1, we observed that no ECG data of normal patients were misidentified, and only 4 out of 31 ECG data of AF patients were incorrectly classified (see Fig. 10). The model achieved a high level of accuracy in Dataset 1, with training and testing accuracy of 99% and 96.55%, respectively. In the case of dataset 1, the model was trained for 50 epochs, however, it was observed that after 30 epochs, the model had reached stability, which is shown in Fig. 8.

However, for dataset 2, 29 out of 105 ECG data of AF patients were misclassified, and 19 out of 131 ECG data of normal patients were also misclassified (see Fig. 11). As a result, Dataset 2 had a higher miss

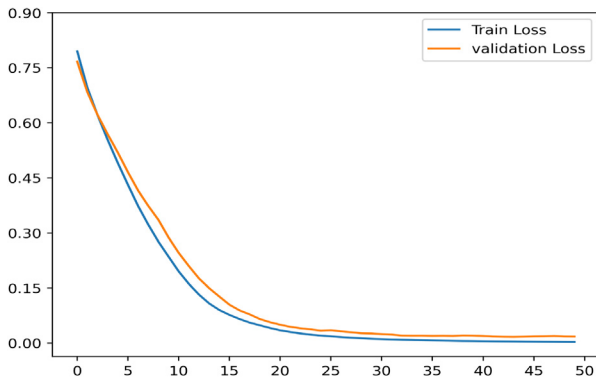


Fig. 12. Training and validation loss curve of proposed AF detection ANN model for dataset 1.

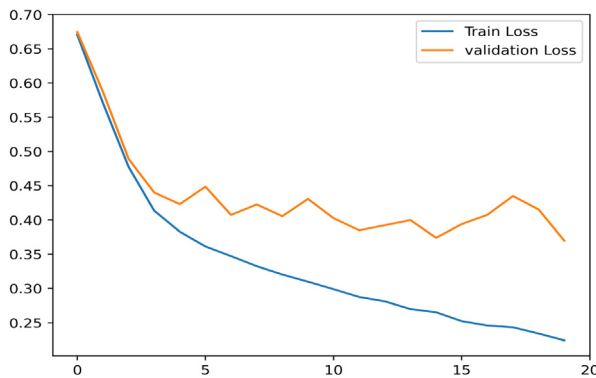


Fig. 13. Training and validation loss curve of proposed AF detection ANN model for dataset 2.

detection rate, with a training and testing accuracy of 91.56% and 86%, respectively. Our proposed model tended to overfit when trained for more than 20 epochs, which is why we chose to limit the number of epochs to 20 during training which is illustrated in Fig. 9.

Based on the accuracy and confusion matrix results, we can conclude that our model is highly reliable for classifying AF and normal patient ECG signals, even with a new set of data. Therefore, we can accurately distinguish between the two types of ECG signals from our dataset. Furthermore, Tables 5 and 6 provide a comparison between the results obtained from our proposed model and the existing related works for both datasets, respectively. In both datasets, the training and testing loss remains very low, which can be seen from Figs. 12 and 13. Additionally, the Area Under the Curve (AUC) summarizes the Receiver Operating Characteristics (ROC) curve, which indicates the classifier's ability to distinguish between different classes. Our proposed model has achieved AUC-ROC values of 0.95 and 0.84 using the First and dataset 2, respectively, as shown in Figs. 15 and 16.

5. Discussion

To ensure the quality of the ECG signal data, we conducted an initial screening process to remove any unreadable data before analyzing it using our neural network algorithms. Our goal was to detect AF by utilizing a new feature extractor that extracts features from a single-lead ECG signal. Our novel feature extractor is designed to efficiently extract effective features from the signal. Prior to feature extraction, the signal is preprocessed using a wavelet transform-based filter that utilizes the 'Sym4' wavelet to remove high-frequency noise. This filter is highly accurate and effectively removes a significant amount of high-frequency noise from the signal.

Table 3

Performance comparison among other ML models and our proposed AF detection model for both datasets.

Model	Accuracy	
	Dataset 1	Dataset 2
SVM	95.96	83.33
Decision tree	94.35	79.45
KNN	97.00	81.97
Logistic regression	95.96	80.81
Proposed model	96.00	86.00

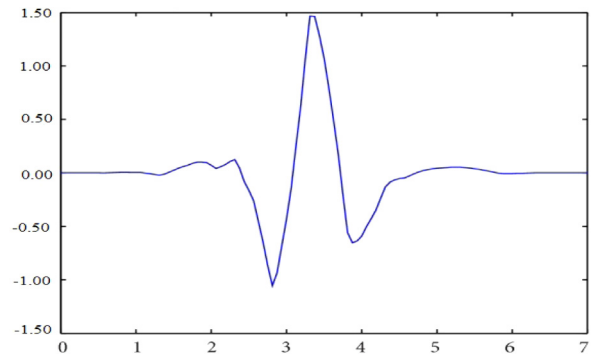


Fig. 14. 'Sym4' wavelet.

After the preprocessing stage, the feature extractor extracts features from the ECG signal. Detecting the R peak is crucial for extracting most of the features accurately. Our novel feature extractor demonstrated high efficiency in R peak detection, which is accomplished through the use of a wavelet transform with the 'sym4' wavelet. This wavelet was selected because its shape is similar to that of the R peak (Fig. 14). Once the R peaks are detected, other features can be extracted using them. The features obtained through this process are highly effective in training the model and detecting AF. Once the model is trained, it can be used to analyze new data and make precise predictions. The most significant aspect of this study was improving the accuracy of AF detection and classification.

The ultimate goal was to achieve an accuracy level as close to 100% as possible, as even a few misdiagnoses could be detrimental. The proposed model demonstrated a precision of 97.5%, recall of 95%, f1-score of 96%, a training accuracy of 99%, and a testing accuracy of 96% for the dataset 1 (Fig. 17). Similarly, for dataset 2, the model achieved a precision of 86%, recall of 84.5%, f1-score of 86%, training accuracy of 91.56%, and testing accuracy of 86% (Fig. 18).

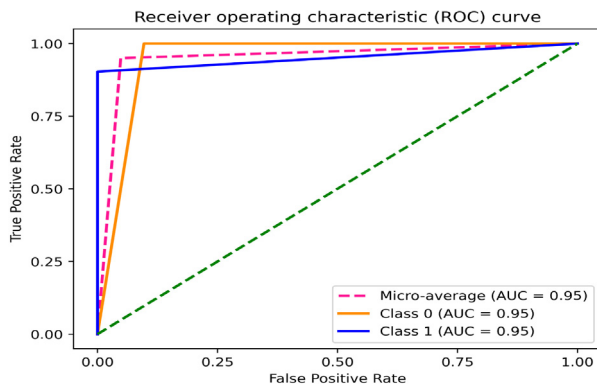
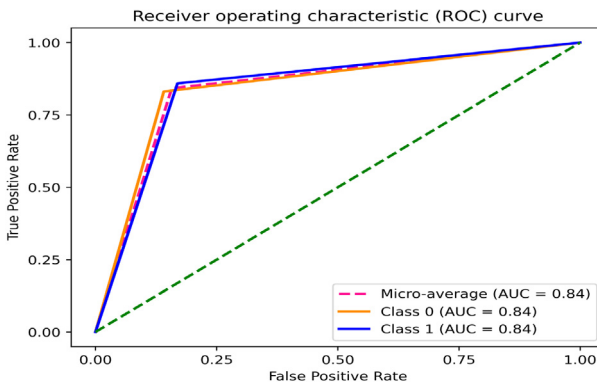
The 5-fold cross-validation of our proposed ANN model on both Dataset 1 and Dataset 2 has yielded highly encouraging results, indicating its strong predictive capabilities. We employed a 5-fold cross-validation technique to validate the model's performance, which resulted in an impressive accuracy rate of 99.75% in Dataset 1 and 90.46% in Dataset 2. These outcomes suggest that our proposed ANN model is capable of accurately predicting the target variable across diverse datasets, demonstrating its adaptability and robustness. The result of cross-validation is shown in Table 7.

The determination of confidence intervals for our proposed detection model has been a critical step in evaluating the accuracy and precision of our model's predictions. By training the model 150 times, we were able to generate a dataset that we used to calculate the 95% confidence interval for both Dataset 1 and Dataset 2. Our analysis revealed that the confidence interval for Dataset 1 was relatively narrow, with an upper limit of 0.9879 and a lower limit of 0.9872. This outcome indicates a high degree of confidence in the accuracy of our model's predictions in this dataset, suggesting that our model is reliable in identifying the target variable in Dataset 1. In contrast, the confidence interval for Dataset 2 was broader, with a lower limit of 0.8652 and an

Table 4

Performance comparison among other existing models and our proposed AF detection model for different Datasets.

Paper	Year	Lead	Dataset	Method	F1- score	Accuracy
Acharya et al. (2017)	2017	Single	MIT-BIH DB	11-layer CNN	–	92.50
Xu et al. (2018)	2018	2 lead	MIT-BIH AFDB	MFSWT and CNN	–	84.85
Xiong et al. (2017)	2017	Single	PhysioNet/CinC 2017	16 layer 1-D CNN	–	82.00
Andersen, Peimankar, and Puthusserypady (2019)	2019	Single	MIT-BIH DB	CNN + LSTM	–	87.40
Attia et al. (2019)	2019	12 lead	Mayo Clinic ECG Laboratory	CNN	0.45	83.30
Cao et al. (2020)	2020	Single	PhysioNet/CinC 2017	LSTM	0.81	78.35
Lai, Bu, Su, Zhang, and Ma (2020)	2020	Multi lead	Hexin Patch Lead II	8-layer CNN	–	93.10
Khriji et al. (2020)	2020	Single	MIT-BIH DB	Deep ANN	–	93.00
Ghiasi et al. (2017)	2017	Single	PhysioNet/CinC 2017	Feature-Based algorithm + DCNN	–	71.00
Baalman et al. (2020)	2020	8 lead	AFACT	R-centered SC-ECG + RNN	0.94	94.00
Chen, Zhang, Liu, Wang, and Wan (2022)	2022	Single	MIT-BIH DB	Feedforward NN	–	84.00
Nguyen et al. (2021)	2021	Single	PhysioNet/CinC 2017	Stacking CNN + SVM	0.83	–
Proposed Model	2023	single	Mendely(MIT-BIH DB)	Novel feature extractor + ANN	0.96	96.00
Proposed Model	2023	single	PhysioNet/CinC 2017	Novel feature extractor + ANN	0.86	86.00

**Fig. 15.** AUC of proposed AF detection ANN model for dataset 1.**Fig. 16.** AUC of proposed AF detection ANN model for dataset 2.

upper limit of 0.8694. This finding suggests less precision and a broader range of possible values for the model's predictions in Dataset 2. It is possible that differences in the data distribution or characteristics of Dataset 2 compared to Dataset 1 contributed to this outcome.

Table 4 presents a comparative analysis of the proposed model and similar works. These models increase the complexity of providing accurate detection results. Our proposed ANN model is a significant binary classifier that is more flexible, less complex, and faster due to its fewer hidden layers. Moreover, our novel feature extractor contributes to faster detection by extracting the most significant features that distinguish AF from normal patient ECG signals. Overall, our proposed model and feature extractor is effective in achieving accurate and efficient detection of AF. Based on the results presented in Tables 5 and 6, our proposed model achieved higher accuracy for Dataset 1 and Dataset 2, respectively. We also evaluate some ML models with these

Table 5

Performance comparison among other existing models and our proposed AF detection model for Dataset 1.

Model	Year	Accuracy
Deep ANN (Khriji et al., 2020)	2020	93.00%
Feedforward NN (Chen et al., 2022)	2022	84.00%
11-layer CNN (Acharya et al., 2017)	2017	94.90%
Proposed Model	2023	96.00%

Table 6

Performance comparison among other existing models and our proposed AF detection model for Dataset 2.

Model	Year	Accuracy
16 layer 1-D CNN (Xiong et al., 2017)	2017	82.00%
LSTM (Cao et al., 2020)	2020	78.35%
DCNN (Ghiasi et al., 2017)	2018	71.00%
Proposed model	2023	86.00%

two datasets and found that our proposed model achieved better accuracy than other ML models which are shown in Table 3. Furthermore, there was no indication of overfitting in our model's accuracy, which suggests that the system is reliable and effective for detecting AF in an unknown dataset. The proposed model performs well for both datasets and achieves better accuracy than other detection models.

One of the advantages of our study is that the proposed model has fewer layers than other detection models, along with an effective feature extractor, reducing complexity and execution time. Additionally, our model can detect and classify both data types accurately, making it applicable to different types of single-lead ECG datasets.

The absence of overfitting in both training and testing results is another significant advantage of our model for achieving reliable and accurate detection. Furthermore, the accuracy of the model can be further increased by using balanced and similar time-segmented datasets.

On the other hand, a limitation of our study is that the model has lower accuracy for Dataset 2 compared to Dataset 1 due to the high low-frequency noise level and dissimilar time periods for each ECG signal. However, this limitation can be addressed by using ECG signals with similar time periods and increasing the sampling rate. Additionally, using imbalanced datasets may reduce the accuracy of the model. It is worth noting that the performance of the proposed model can be improved for properly shaped single-lead ECG signals.

5.1. Future research

In the future, we will upgrade our feature extractor to extract features more accurately. Our main focus will be developing a feature extractor and detection model which will work better and more accurately with a noisy signal. We will also work to develop a model which can detect AF from multi-lead ECG signals and is also user-friendly and easy to use.

Table 7
Result of 5 fold cross-validation and Confidence interval.

Folds	Dataset 1			Dataset 2		
	Acc. of each fold	Avarage	Confidence interval (At 95%)	Acc. of each fold	Average	Confidence interval (At 95%)
1st	0.98			0.86		
2nd	1.00			0.87		
3rd	1.00	0.99	0.9872–0.9879	0.93	0.90	0.8652–0.8695
4th	1.00			0.88		
5th	1.00			0.95		

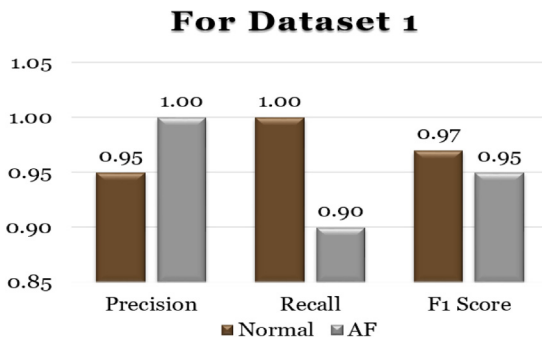


Fig. 17. Performance analysis graph of proposed AF detection ANN model for dataset 1.

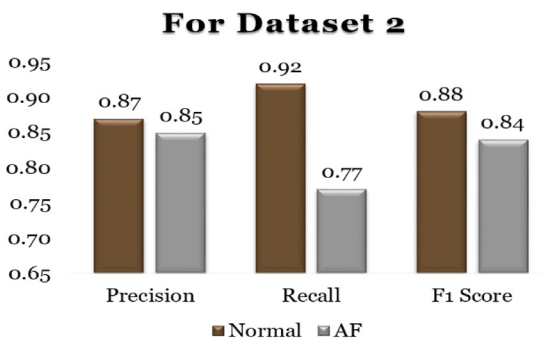


Fig. 18. Performance analysis graph of proposed AF detection ANN model for dataset 2.

6. Conclusions

In summary, AF is a significant cardiac arrhythmia that can lead to fatal outcomes if not detected and treated at an early stage. In this study, we propose a novel feature extraction method and a DL-based classification model for AF detection. The feature extractor with the conjunction of the proposed model demonstrated exceptional performance, achieving the accuracy of 96% and 86% on two distinct datasets, respectively. Additionally, employing 5-fold cross-validation, the model achieved impressive accuracy rates of 99% for Dataset 1 and 90% for Dataset 2, respectively. The high AUC values obtained from the proposed model indicate the strong ability of the model to differentiate AF from non-AF cases, emphasizing its reliability and stability across different datasets and validation techniques. The findings of this study have significant implications for healthcare professionals as it provides a reliable and early detection tool for AF. By accurately identifying AF cases, timely interventions and appropriate treatments can be initiated, ultimately improving patient outcomes and reducing the risks of severe complications such as stroke and cardiac arrest. In conclusion, this study offers a promising solution for the accurate and early diagnosis of AF, which can lead to better treatment outcomes and decrease the chances of fatal diseases such as stroke and cardiac arrest. The study highlights the importance and effectiveness of DL techniques in AF detection and encourages further research in this domain.

CRediT authorship contribution statement

D.U.S. Duranta: Conceptualization, Analysis, Methodology, Software development, Validation Writing of the original draft. **Md Shahin Ali:** Data curation, Analysis, Methodology, Reviewing, and editing of the manuscript. **Abhilash Arjan Das:** Reviewing and editing of the manuscript. **Md Mahbubur Rahman:** Reviewing and editing of the manuscript. **Md Manjurul Ahsan:** Reviewing and editing of the manuscript. **Md Sipon Miah:** Data curation, Software development, Reviewing of the manuscript. **Md Khairul Islam:** Supervision, Reviewed and editing of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request

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